

MDM

Q.1) Draw and explain the architecture of 8086 microprocessor.

Q.2) Explain various addressing modes of 8086 microprocessor.

Q.3) Explain in detail with diagram timers of 8051 microcontroller. (SCON, TCON, TMOD register comes under this question and they ask single register for 5M and for 10M complete question)

Q.4) Explain various operating modes of 8255 PPI.

Q.5) Explain serial communication in 8051 using SCON register.

Q.6) Explain memory organization of 8051 microcontroller

Q.7) Explain various addressing modes of 8051 microcontroller.

Q.8) Draw and explain the architecture of Pentium processor with neat diagram.

Q.9) Explain the interfacing of a stepper motor to 8051 microcontroller.

Q.10) State and explain the features of Pentium processor and ARM cortex.

Q.11) Explain the interrupt structure of the 8086 processor (IVT) and differentiate between

Hardware and software interrupts.

Q.12) Define assembler directives. Explain all the assembler directives supported by 8086

microprocessors with examples.

Q.13) Draw and explain read and write operation timing diagram for 8086 microprocessor in

minimum and maximum mode

Q.14) Draw and explain the architecture of ARM cortex core.

Q.15) Draw and explain the Program status word (PSW) register of 8051.

Q.16) Explain the following instruction of 8086 microprocessor: STOSB, DAA, CALL, XLAT, LAHF, AAA, XLATB, DIV AX and CMPSB.

Q.17) Explain the generation of the clock and reset signals using the 8284 clock generator.

Q.18) Draw and explain the interfacing of a LCD to the 8051.

Q.19) Explain memory segmentation of 8086 microprocessor. (Note: This question may be directly

asked as logical and physical address in 8086, as they are part of memory segmentation)

Q.20) Draw and explain the architecture of 8051 microcontroller in detail.

Q.21) Explain the modes of 8255 with a neat block diagram.

Q.22) Explain the internal RAM organization of the 8051.

Highlight the importance of the register

Banks.

Q.22) Differentiate between the following:

- i) Microprocessor and Microcontroller
- ii) Stepper motor and Dc motor interfacing
- iii) Minimum and maximum mode (Note may this question ask in explanation form also)

Q.23) Write a short note on the following:

- i) Function of port 0 of 8051
- ii) Interfacing of DC motor to 8051 microcontroller
- iii) Memory banking in 8086.
- iv) String instruction of 8086 and Arithmetic instruction of 8051 microcontroller
- v) port structure of the 8051 microcontroller.

vi) pipeline features of 8086 in detail.

Programming

Q.1) Write a program in 8086 to exchange contents of two memory blocks.

Q.2) Write an assembly language program to arrange the 10 number in ascending order using 8086 microprocessor.

Q.3) Write a program to transfer block of data (10 bytes) from DS: 2000H to ES: 4000H using instruction set of 8086.

Q.4) Write a program to generate a square wave on Port Pin 2.0 using Timer 0 in Mode 1 in 8051.

Q.5) Write an assembly language program using 8051 to swap two numbers.